# Observability-Aware Trust-Based Grayhole Detection for Underwater Acoustic Networks

CS301 Computer Networks Mini Project, Team 9


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# Problem statement

- In an underwater acoustic network, sensors send data to the sink through relays (network layer).
- A **grayhole** relay drops a part of the packets that it must forward (here 30%). Its loss looks like normal link loss.
- The **watchdog** listens for each forward and accuses a relay with too many misses.

**A forward is missing when**

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| the relay **dropped** it (the attack) | the monitor was **transmitting** (half-duplex) | the packet **collided** at the relay | the **long link** lost it |

Only cause 1 is an attack. Causes 2 to 4 increase with traffic, so a fixed threshold **accuses honest relays**.

# Proposed approach

## 1. Observability filter

A monitor counts a watch window only if it could hear the forward: it did not transmit, the relay did not transmit, and no known frame arrived at the relay.

## 2. Expected forwards

Each link has an expected loss from its distance and the median loss of other links of similar length.

$$\text{Trust}_B = \frac{\sum \text{heard}}{\sum \text{expected}}$$

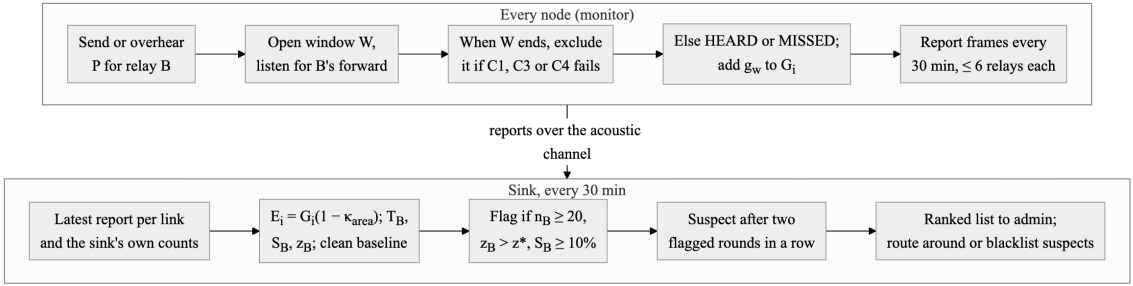
## 3. Decision at the sink

Dynamic threshold, baseline cleaning, and a suspect only after two flagged 30-min rounds in a row.

## Improvement over the baselines

The fixed watchdog counts every silence as a drop. The CAD-style detector uses an average collision rate. Our detector excludes each window that the monitor could not observe, and needs no probe packets.

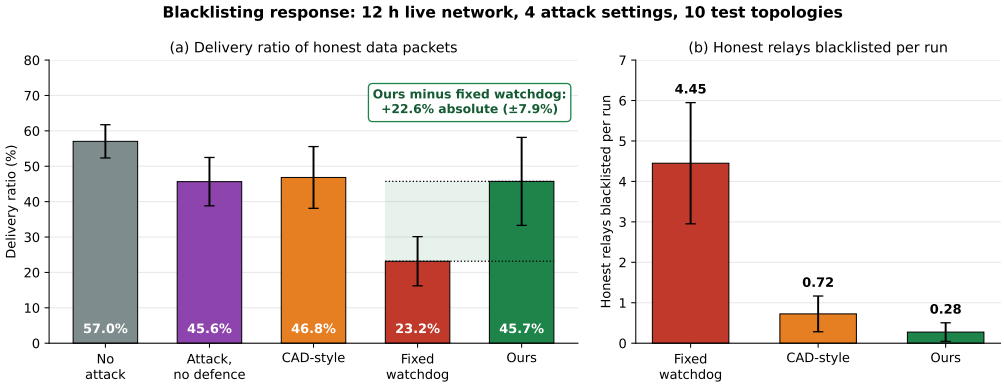
# Architecture




| | |
|---|---|
| Simulator | UnetStack 7.0.0 (Groovy agents), 20 sensors and one sink, 4 km × 4 km |
| Classes | `MonitorNode`, `Detectors`, `Topology`, `DistanceLossChannel` |
| Fair test | Thresholds tuned on topologies 101 to 106, tested on topologies 201 to 210 |

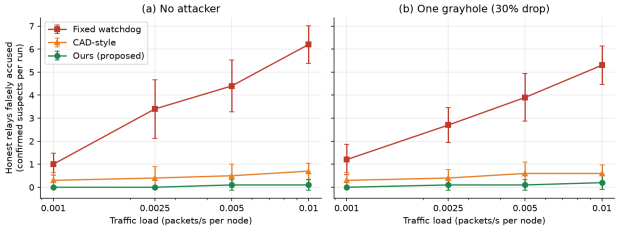
# Results: delivery ratio with blacklisting

**Delivery ratio** = data packets from honest sensors that reach the sink ÷ data packets they generate.




With the fixed watchdog, the delivery ratio is **23.2%**. With ours, it is **45.7%**.
Absolute increase: **22.6%** (±7.9%), because ours blacklists 0.28 honest relays per run and the fixed watchdog 4.45.

# Results: false accusations




Honest relays accused per run as the traffic load changes

**Live network, 12 h**

| Method | Caught | Honest relays accused per run |
|---|---|---|
| Fixed watchdog | 100% | 7.2 |
| CAD-style | 58% | 0.8 |
| **Ours** | **95%** | **0.3** |

# Conclusion

- Under water, a missing forward is often not an attack.
- Our detector judges only the windows that the monitor could observe, and compares them with the expected loss of each link.
- It caught 95% of the grayholes and accused 0.3 honest relays per run. The fixed watchdog accused 7.2.
- With blacklisting, it kept a delivery ratio of 45.7%. The fixed watchdog kept 23.2%.
- Future work: dishonest monitors, mobile nodes, and learning the loss curve during operation.

**Thank you**